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Senior Software Engineer with 18+ years building scalable backend systems, abuse prevention at planetary scale (YouTube/Google), cloud infrastructure, high-performance data pipelines, NLP/AI engines, and mobile apps. Expertise in Go, distributed systems, performance optimization, spam/malware detection, and reliable production software.

Skills Languages: Go, Python, Dart Technologies: Google Cloud, App Engine, Postgres, Pub/Sub, NLP/AI systems, iOS/Android mobile, 3D algorithms, abuse/spam detection Other: Distributed systems, performance tuning, automated testing

#### Professional Experience

Independent Contractor / Multiple Engagements Dec 2019 – Present (Remote) illumineX (Software Engineer, Nov 2021 – Present) • Rewrote core NLP engine for mīstro AI trading system, boosting accuracy/throughput significantly and adding 1,000+ automated test cases for 95%+ coverage • Designed and launched first.then.next — AI-powered calendar system to promote user independence and goal achievement

YouLi (youli.io) (Mobile App Contractor) • Built YouLi Trip mobile app (iOS/Android) with full offline trip page support, enabling seamless access for users in low/no connectivity

Stairwell (Software Engineer) Apr 2021 – Oct 2021 (Remote) • Developed new malware ingester pipeline, expanding detection capabilities for cloud-scale threat intelligence

Speakeasy (Cofounder & Engineer) Apr 2020 – Apr 2021 (Remote) • Architected full-stack video events platform supporting 300+ concurrent participants reliably • Collaborated with cofounders to secure first paying customer, running multiple high-attendance weekly events

Google Apr 2016 – Mar 2019 (San Francisco, CA) Go Cloud Team (Aug 2017 – Mar 2019) • Designed and released portable Pub/Sub API, enabling cross-cloud/event-driven architectures

App Engine Runtimes Team (Apr 2016 – Aug 2017) • Launched PHP 7.2 runtime (beta), adopted widely (Hacker News #1 discussion); improved developer experience for legacy PHP apps

YouTube (Software Engineer) Dec 2008 – Oct 2010 (San Bruno, CA) • Built dependency tracking system for efficient denormalized view generation in sharded data warehouse • Implemented atomic updates across sharded tables, ensuring consistency at massive scale • Created Sawzall-based abuse detection suite and stealth features neutralizing mass spam accounts without alerting attackers • Eliminated chain comments exploit and added robust filters for comment/message

spam, drastically reducing abuse vectors • Developed brute-force password attack mitigations, enhancing account security

VideoSurf (Software Engineer) Oct 2010 – Sep 2012 (San Mateo, CA) • Optimized text-mining/deduplication pipeline, accelerating per-video processing from seconds to 50 ms (40–100x speedup) • Leveraged advanced Postgres indexing to enable thousands-fold faster catalog updates, allowing daily full-site refreshes • Built automated data quality reports and daemons for duplicate video detection via fuzzy matching

Early Career Summary Google (Support Engineer, Ad Spam Operations) 2007–2008 • Developed phone verification system reducing failures to <25% of legacy • Built GWT UI for spam investigations UC Davis (Programmer IV) 2002–2006 • Created 3D brain visualization tools Idol Minds (Programmer) 2001–2002 • Adapted mesh simplification algorithms for PS2 game rendering NASA Ames (Intern) 1999 • Implemented tetrahedral mesh simplification with scalar fields

Education Brown University – M.Sc., Applied Mathematics (1999–2000) University of California, Davis – B.S., Mathematics (1994–1999)